



P H A N T O M

blindspot

*detecting what the
tracker cannot see*

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The Problem

Perry update here

What We Are Building

Perry update here

Why It Matters

Perry update here

Christine · Student Researcher · Austin Peay State University

PHANTOM

Poisoning and Detection of Hidden Embedding Attacks
in Multi-Camera Object Tracking

What I have learned	What I am working on	Where this is going
How ByteTrack and BoT-SORT-ReID link object identities across frames using appearance embeddings Put more info here Put more info here Put more info here	Watching through NDSU intersection scenarios S01–S09 to find the best 2-minute trim windows Finding 2–3 independent papers each week on multi-camera tracking and ReID benchmarks Put more info here Put more info here	Put info here Put info here Put info here Put info here



Floyd · Student Researcher · Austin Peay State University

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Poisoning and Detection of Hidden Embedding Attacks
in Multi-Camera Object Tracking

What I have learned

- How pixel-level adversarial perturbations destroy tracker confidence heatmaps and cause Kalman filter drift
- How a single camera can mess with an entire city's global Re-ID model, as well as allow someone to walk as a ghost through any camera sharing the database
- How AI/ML models can create custom pixels to mess with a camera's tracking, as well as make it invisible to the human eye and bypass security software.
- Methods of how data poisoning works, as well as how easy it is to trick a camera's feed without the camera realizing, using adversarial attacks, backdoor triggers, or changed training data

What I am working on

- Watching through NDSU intersection scenarios S10–S18 to find the best 2-minute trim windows
- Finding 2–3 independent papers each week on adversarial ReID and embedding-space attacks
- Creating powerpoints over said papers to present and explain my understanding of the papers and what their arguments and methods are
- Learning Python so that I can better understand what we are looking for in code work.

Where this is going

- Hoping that by the end I have a full Understanding of Data poisoning methods
- Learning and understanding how backdoor triggers work, and hopefully finding a way to find them and quarantine said triggers.
- Learn how Python works fully, and be able to understand and change code so that it runs for my area of work.
- Create a software or figure out how to remove backdoored, or poisoned Re-ID models, and detect what the tracker cannot see.

1-minute talk · Responsible for scenarios S10–S18 · Threat model and poisoning experiment